1. Sequence : 4, 11, 18, 25……
   A. Find the general term of the given sequence.
   B. What is the sum of first 25 terms of this sequence?
   C. What will be the sum of the last 15 terms of the first 25 terms?

2. (a)Sequence : 2, 7, 13, 20……
   (b)Sequence : 1, 8, 16, 25….
   A. Find the general expression to generate nth term of sequence a.
   B. Find sum of first 15 terms of sequence b.
   C. Find 101 th term of sequence b.

3. **Sequence:**
   5,9,13,17,…
   Questions
   A. Find the **general term ana_nan** of the sequence.
   B. Find the **sum of the first 30 terms**.
   C. Find the **sum of the last 10 terms among the first 30 terms**.

4. **(a) Sequence:**
   3,6,10,15,21,…
   (b) Sequence:
   2,6,12,20,30,…
   A. Find the **general expression for the nnn-th term of sequence (a)**.
   B. Find the **sum of the first 12 terms of sequence (b)**.
   C. Find the **50th term of sequence (b)**.

5. **Sequence:**
   7,14,21,28,…
   A. Find the **general term of the sequence**.
   B. Find the **sum of the first 40 terms**.
   C. Find the **sum of the last 12 terms among the first 40 terms**.

6. **(a) Sequence:**
1,4,9,16,25,…

(b) Sequence:
3,9,27,81,…

A. Find the **general term of sequence (a)**.
B. Find the **sum of the first 8 terms of sequence (b)**.
C. Find the **12th term of sequence (b)**.

7. A rectangular pond has length of 8m, width of 6m, depth of 3m.
A. Find its volume.
B. If we were to cover the inside of the pond with wood slabs with negligible thickness, what will be the area of the slabs required to cover ?
C. suppose we are going to build a road across the pond with width of 3m. if we need 7.5 taka to cover each m^2 area of the road then what will be the total cost to cover the road?

8. A **rectangular field** has a **length of 60 m** and a **width of 40 m**.
Two roads of **equal width 2 m** cross each other **exactly at the center** of the field — one along the length and the other along the width.

Questions

A. Find the **area of the road along the length**.

B. Find the **area of the road along the width** (excluding the overlapping square in the middle).

C. Find the **total area occupied by the two roads**.

9. A **rectangular iron block** has:

- Length = **10 cm**
- Width = **7 cm**
- Height = **3 cm**

The block is placed inside a **rectangular container** having:

- Length = **20 cm**
- Width = **10 cm**
- Height = **6 cm**

The container is filled completely with water.
Iron is **7.8 times heavier than water**.

Questions

A. Find the **volume of the container**.

B. Find the **weight of the iron block**.

C. If the iron block is removed while the container is full of water, find the **new height of the water in the container**.

10. A **rectangular metal block** has:

- Length = **12 cm**
- Width = **5 cm**
- Height = **4 cm**

It is placed in a **rectangular water tank** having:

- Length = **18 cm**
- Width = **12 cm**
- Height = **8 cm**

The tank is filled with water.
The density of the metal is **6 times the density of water**.

Questions

A. Find the **volume of the tank**.

B. Find the **weight of the metal block**.

C. If the block is removed from the full tank, determine the **new height of the water level**.

11. The cost of covering the **floor of a rectangular hall with marble tiles** is **9600 taka** at a rate of **20 taka per m²**.

If the **length were increased by 4 m** and the **width decreased by 2 m**, the cost would become **9800 taka** at the same rate.

Questions

A. Find the **original length and width** of the hall.
B. Find the **area of the hall**.
C. If the rate increases to **25 taka per m²**, find the **new total cost**.

12. A **rectangular room** is covered with carpet costing **15 taka per m²** and the total cost is **6750 taka**.

If the **length is 3 m more than the width**, and the **perimeter of the room is 60 m**:

Questions

A. Find the **length and width of the room**.
B. Verify the **total area of the floor**.
C. If the carpet price rises to **18 taka per m²**, find the **new cost to cover the floor**.

13. The cost of covering the **floor of a rectangular classroom with wooden boards** is **12,000 taka** at **30 taka per m²**.

If the **length is reduced by 5 m** and the **width increased by 2 m**, the cost becomes **10,800 taka** at the same rate.

Questions

A. Find the **original length and width of the classroom**.
B. Find the **original floor area**.
C. If the floor must be polished at **5 taka per m²**, find the **additional polishing cost.**

14. Two numbers a and b satisfy a>b. The sum of the numbers is **14** and their product is **45**.

a) Factorize using the identity method: $(3x+5)(3x-2)$

b) Find the value of: $3a^2+3b^2$

c) Prove that: $(a+3b)^2-8b^2=364$

15. Two numbers a and b satisfy a>b. The sum of the numbers is **18** and the product is **72**.

a) Express the given expression as subtraction of two squares:

$(5x+4)(5x-6)$

b) Find the value of:

$2a^2+2b^2-ab$

c) Prove that:

$(a+3b)^2-5ab=486$

16. 2 numbers a and b satisfies a>b. a + b = 12, ab = 32

a) Find value of $a^2 - b^2$

b) Find value of $a^3 + b^3$

c) Suppose x + y = 20, xy = 96, show $(x+2y)^2 + (x - y)^2 = 656$

17. $\frac{1}{x} + x = 2\,, m - \frac{1}{m} = 7$

a) find $x - \frac{1}{x}$.

b) find $x^2 + \frac{1}{x^2}$.

c) find $m^2 - \frac{1}{m^2}$.

18. $\frac{1}{x} - x = 4, m + \frac{1}{m} = 9$

a) find $x + \frac{1}{x}$.

b) find $x^2 + \frac{1}{x^2}$.

c) find $m^4 - \frac{1}{m^4}$.

19. **$5x - 3y = 1$**, $5x + 3y = 19$(you aren't allowed to find direct value of x and y via solving equation, rather use algebraic formulas to find answer)

a) find value of 30xy.

b) find value of $25x^2 - 9y^2$

c) find value of $(10x - 6y + 7)(10x + 6y + 10)$

20. $3x - 5y = 4$, $30xy = 90$(you aren't allowed to find direct value of x and y via solving equation, rather use algebraic formulas to find answer)

a) find value of $9x^2 + 25y^2$.

b) find value of $3x + 5y$.

c) find value of $(9x - 15y + 8)(9x + 15y - 4)$

21. $4x - 3z = 0$, $24xz = 288$(you aren't allowed to find direct value of x and y via solving equation, rather use algebraic formulas to find answer)

a) find value of $9z^2 - 16x^2$.

b) find value of $3z + 4x$.

c) find value of $(16x - 12z + 31)(16x - 12z + 9)$

22. **i.** a book has 200 pages, each with thickness of .2 mm

ii. a cubic pool can store maximum $4096m^3$ amount of water

a)find length of each side of the pool

b) if the book has length of 30cm and width of 8 inch, find its volume in m^3 unit.

c) if we try to paint the pool, a we need 30tk to paint per square meter, what will be the total cost?

23. a cubic pool can carry maximum 6000L of water. But there is a cubic metal of length 85 inch, which is 5 times heavier than water.

a) find length of each side of the pool in meter unit.

b) if we were to fill the whole pool with the metal, what will be the metal's total weight.

c) if we throw out the metal from the pool after it was filled with water, find the decrease in height of the water level.

24. Convert following amounts to specified unit

a)24.56m^3 -> L , b) 4.67 L -> cm^3, c) 5.67 inch -> m, d) 43.90 cm -> foot, e) 4.67mile -> km

f) 432.9 km -> mile, g)329.456 km -> nautical mile, h) 54.31 cm^3 -> L, i)98.35 nautical mile -> km

j) 326.54 L -> m^3